

Lab 1 ticket

Finding Similar Things — Friday, September 18 (short)

Name: _____ Language (circle): Octave / Python

New syntax

I needed to...	Command / pattern
Distance / norm	
Find the minimum	

Data: $x = (2, 1)$; $z_1 = (2, 2)$, $z_2 = (0, 1)$, $z_3 = (3, 1)$, $z_4 = (2, 0)$, $z_5 = (1.5, 1.5)$.

E1. Distances $\|x - z_i\|$ and nearest neighbor.

Expected: 1, 2, 1, 1, $\sqrt{1/2}$; nearest z_5 .

Yours: $d =$ _____

Nearest: z _____

E2. Multiply every first coordinate by 10. New nearest neighbor(s)? Changed?

Expected: tie z_1 and z_4 ; yes, changed.

Yours: nearest _____ changed? yes / no

One sentence why scale matters: _____

Instructor initials: _____